Lab 6

Formation of Scrum Team, Product Backlog Creation, and High-Level Sprint Planning

Part A – Scrum Team Identification

Task A1: Assign Roles

Role	Team Member Name	Responsibilities
Product Owner	Nikhil Savanth K M	• Represents the customer • Prioritizes product backlog • Defines acceptance criteria
Scrum Master	Ullas Gowda R	• Ensures Scrum practices are followed • Facilitates meetings • Removes blockers
Developer	Shreyas C P	• Implements product features • Designs, develops, and tests
Developer	Siddeshwarprasad K R	• Implements product features • Designs, develops, and tests

Part B – Product Backlog Creation

Task B1: Write User Stories

1. As a user, I want to register an account so that I can access the system and place orders.
2. As a user, I want to log in securely so that I can access my personal account.
3. As a user, I want to log out so that my account remains secure after use.
4. As a user, I want to browse products by category so that I can easily find items.
5. As a user, I want to search for products so that I can quickly find specific items.
6. As a user, I want to view product details so that I can make informed purchase decisions.
7. As a user, I want to add products to my cart so that I can purchase multiple items together.
8. As a user, I want to remove items from my cart so that I can modify my order before checkout.
9. As a user, I want to update product quantity in my cart so that I can control my purchase amount.
10. As a user, I want to place an order so that I can buy selected products.
11. As a user, I want to make secure online payments so that my transactions are safe.
12. As a user, I want to view my order history so that I can track my past purchases.
13. As an admin, I want to add new products so that users can browse and purchase them.
14. As an admin, I want to update or delete products so that product information remains accurate.
15. As an admin, I want to manage customer orders so that I can ensure proper order processing.

Task B2: Product Backlog Table

Backlog ID	User Story	Priority	Story Points
PB-1	User registration	High	3
PB-2	User login	High	3
PB-3	Browse products by category	High	5
PB-4	View product details	High	3
PB-5	Add to cart	High	5
PB-6	Place order	High	8
PB-7	Make payment	High	8
PB-8	Search products	Medium	5
PB-9	Remove items from cart	Medium	3
PB-10	Update cart quantity	Medium	3
PB-11	View order history	Medium	3
PB-12	User logout	Medium	1
PB-13	Admin add product	Low	5
PB-14	Admin update/delete product	Low	5
PB-15	Admin manage orders	Low	5

Part C – High Level Sprint Plan

Task C1: Define Sprint Goals

Sprint	Sprint Goal
Sprint 1	Set up project and implement user registration and login functionality
Sprint 2	Implement product browsing and product details viewing
Sprint 3	Implement shopping cart functionality (add/remove/update items)
Sprint 4	Implement order placement and checkout process
Sprint 5	Integrate payment system and develop admin product management features
Sprint 6	Perform system testing, bug fixing, and performance optimization

Task C2: Sprint Planning Table

Sprint	Backlog Items	Expected Deliverables
Sprint 1	PB-1 User Registration, PB-2 User Login	Authentication module (user can register and log in)
Sprint 2	PB-3 Browse Products, PB-4 View Product Details	Product browsing and product details functionality
Sprint 3	PB-5 Add to Cart, PB-9 Remove from Cart, PB-10 Update Cart Quantity	Shopping cart module
Sprint 4	PB-6 Place Order	Order processing module
Sprint 5	PB-7 Make Payment, PB-13 Admin Add Product, PB-14 Admin Manage Products	Payment integration and admin product management
Sprint 6	PB-8 Search Products, PB-11 Order History, PB-12 Logout, PB-15 Admin Manage Orders	Final system with additional features, testing, and stable release

Conclusion

In this lab, the Scrum framework was applied by forming a Scrum team with clearly defined roles such as Product Owner, Scrum Master, and Development Team. A comprehensive Product Backlog was created by converting system features into user stories and prioritizing them based on business value. Furthermore, a high-level sprint plan was developed to distribute backlog items across multiple sprints, ensuring incremental and structured development. This lab provided practical understanding of Scrum practices, backlog management, and sprint planning, highlighting how Agile methodologies enable efficient and flexible software development.

Team Members

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